

INFLUENCE OF MACH NUMBER AND 3D INSTABILITIES ON THE TRANSITION MECHANISMS OF LAMINAR BOUNDARY LAYERS OVER GAP DISCONTINUITIES

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In this work we investigate the linear stability of laminar boundary layers encountering a gap discontinuity. The gap is modeled as a rectangular cavity of width w and depth d embedded in a flat plate aligned with the flow. The depth of the cavity is varied within the range $d/\delta^* \in [0.5, 4]$, while the width is set to $w/\delta^* \in [5, 100]$. The Reynolds number considered is $Re_{\delta^*} = 1000$, where δ^* denotes the displacement thickness measured at the upstream edge of the gap on a smooth surface that is free of discontinuities. High-fidelity simulations for both incompressible and compressible flows are performed using the spectral/ hp element method implemented in Nektar++ [1]. Under incompressible or low-Mach-number conditions, previous experimental studies [2] have shown that narrow gaps amplify Tollmien–Schlichting (TS) waves, leading to transition downstream through the primary instability pathway, as illustrated in Figure 1. In contrast, sufficiently wide gaps promote a bypass transition originating near the downstream edge of the gap. In incompressible simulations, for fixed gap depth d , a Hopf bifurcation emerges as the gap width exceeds a critical value $\mathcal{H}(d)$. The periodic orbit emerges as a form of self-sustained oscillation near the downstream edge of the gap. The critical width for this bifurcation increases as the gap depth decreases and disappears for shallow gaps ($d/\delta^* < 1.25$), which remain globally stable. The functions $d \mapsto \mathcal{B}(d)$ (bypass threshold) and $d \mapsto \mathcal{H}(d)$ show good agreement. We further extend the analysis to examine the effects of Mach number ($M \in (0, 1)$) and three-dimensional gap geometries on the bifurcation behavior. Results indicate that increasing Mach number lowers the critical width $\mathcal{H}(d)$, suggesting that compressibility promotes bypass transition.

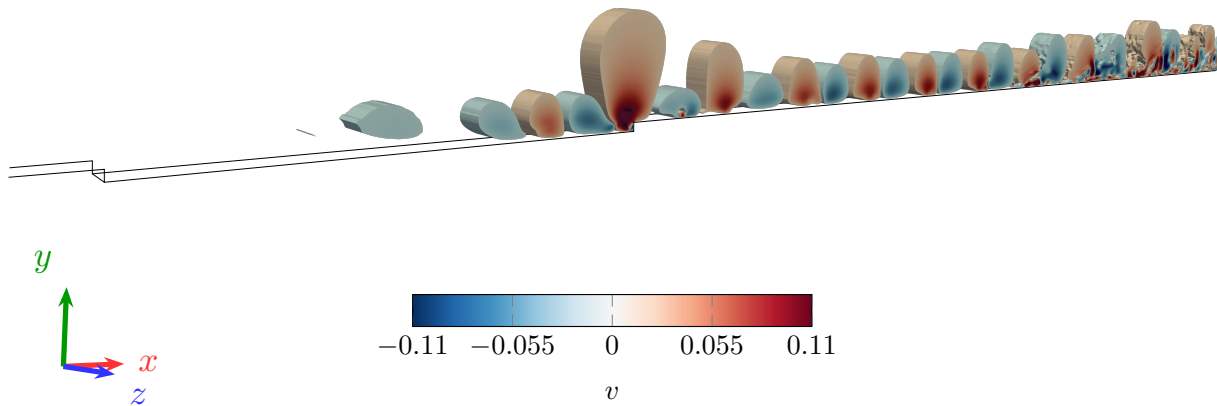


FIGURE 1. Wall-normal velocity isovolumes showing the over-amplification of TS waves due to the presence of a 3D gap discontinuity of depth $d = 1.5\delta^*$ and width $w = 80\delta^*$, and transition to turbulence downstream of the gap.

References

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- [2] J. D. Crouch, V. S. Kosorygin, M. I. Sutanto, and G. D. Miller. Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability. *Flow*, 2:E8, 2022. DOI: 10.1017/f1o.2022.1.